

sysy22

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Chapter 1

Namespace Index

1.1 Namespace List

Here is a list of all namespaces with brief descriptions:

frontEnd	9
frontEnd::ast	9

Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

Sysy22BaseVisitor	
ASTVisitor	11

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

ASTVisitor	11
--------------------------------------	----

Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

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Chapter 5

Namespace Documentation

5.1 frontEnd Namespace Reference

Namespaces

- namespace [ast](#)

5.2 frontEnd::ast Namespace Reference

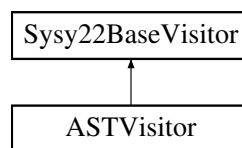
Chapter 6

Class Documentation

6.1 ASTVisitor Class Reference

```
#include <ASTVisitor.h>
```

Inheritance diagram for ASTVisitor:



The documentation for this class was generated from the following file:

- src/frontend/[ASTVisitor.h](#)

Chapter 7

File Documentation

7.1 src/frontend/AST.cpp File Reference

7.2 src/frontend/AST.hpp File Reference

Namespaces

- namespace [frontEnd](#)
- namespace [frontEnd::ast](#)

7.3 AST.hpp

[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 namespace frontEnd {
00004     namespace ast {
00005
00006
00007
00008     };
00009 };
00010
```

7.4 src/frontend/ASTVisitor.cpp File Reference

7.5 src/frontend/ASTVisitor.h File Reference

```
#include "grammar/Sysy22BaseVisitor.h"
```

Classes

- class [ASTVisitor](#)

7.6 ASTVisitor.h

[Go to the documentation of this file.](#)

```

00001 #pragma once
00002
00003 #include "grammar/Sysy22BaseVisitor.h"
00004
00005 class ASTVisitor : public Sysy22BaseVisitor {
00006
00007     virtual std::any visitProg(Sysy22Parser::ProgContext *ctx) override {
00008         return visitChildren(ctx);
00009     }
00010
00011     virtual std::any visitCompUnits(Sysy22Parser::CompUnitsContext *ctx) override {
00012         return visitChildren(ctx);
00013     }
00014
00015     virtual std::any visitCompUnit(Sysy22Parser::CompUnitContext *ctx) override {
00016         return visitChildren(ctx);
00017     }
00018
00019     virtual std::any visitDecl(Sysy22Parser::DeclContext *ctx) override {
00020         return visitChildren(ctx);
00021     }
00022
00023     virtual std::any visitConstDecl(Sysy22Parser::ConstDeclContext *ctx) override {
00024         return visitChildren(ctx);
00025     }
00026
00027     virtual std::any visitConstDef(Sysy22Parser::ConstDefContext *ctx) override {
00028         return visitChildren(ctx);
00029     }
00030
00031     virtual std::any visitVarDecl(Sysy22Parser::VarDeclContext *ctx) override {
00032         return visitChildren(ctx);
00033     }
00034
00035     virtual std::any visitVarDef(Sysy22Parser::VarDefContext *ctx) override {
00036         return visitChildren(ctx);
00037     }
00038
00039     virtual std::any visitInit(Sysy22Parser::InitContext *ctx) override {
00040         return visitChildren(ctx);
00041     }
00042
00043     virtual std::any visitInitList(Sysy22Parser::InitListContext *ctx) override {
00044         return visitChildren(ctx);
00045     }
00046
00047     virtual std::any visitInt(Sysy22Parser::IntContext *ctx) override {
00048         return visitChildren(ctx);
00049     }
00050
00051     virtual std::any visitFloat(Sysy22Parser::FloatContext *ctx) override {
00052         return visitChildren(ctx);
00053     }
00054
00055     virtual std::any visitFuncDef(Sysy22Parser::FuncDefContext *ctx) override {
00056         return visitChildren(ctx);
00057     }
00058
00059     virtual std::any visitFuncType(Sysy22Parser::FuncTypeContext *ctx) override {
00060         return visitChildren(ctx);
00061     }
00062
00063     virtual std::any visitFuncFParams(Sysy22Parser::FuncFParamsContext *ctx) override {
00064         return visitChildren(ctx);
00065     }
00066
00067     virtual std::any visitScalarParam(Sysy22Parser::ScalarParamContext *ctx) override {
00068         return visitChildren(ctx);
00069     }
00070
00071     virtual std::any visitArrayParam(Sysy22Parser::ArrayParamContext *ctx) override {
00072         return visitChildren(ctx);
00073     }
00074
00075     virtual std::any visitBlock(Sysy22Parser::BlockContext *ctx) override {
00076         return visitChildren(ctx);
00077     }
00078
00079     virtual std::any visitBlockItem(Sysy22Parser::BlockItemContext *ctx) override {
00080         return visitChildren(ctx);
00081     }
00082

```

```

00083     virtual std::any visitAssign(Sysy22Parser::AssignContext *ctx) override {
00084         return visitChildren(ctx);
00085     }
00086
00087     virtual std::any visitExprStmt(Sysy22Parser::ExprStmtContext *ctx) override {
00088         return visitChildren(ctx);
00089     }
00090
00091     virtual std::any visitBlockStmt(Sysy22Parser::BlockStmtContext *ctx) override {
00092         return visitChildren(ctx);
00093     }
00094
00095     virtual std::any visitIfElse(Sysy22Parser::IfElseContext *ctx) override {
00096         return visitChildren(ctx);
00097     }
00098
00099     virtual std::any visitWhile(Sysy22Parser::WhileContext *ctx) override {
00100         return visitChildren(ctx);
00101     }
00102
00103     virtual std::any visitBreak(Sysy22Parser::BreakContext *ctx) override {
00104         return visitChildren(ctx);
00105     }
00106
00107     virtual std::any visitContinue(Sysy22Parser::ContinueContext *ctx) override {
00108         return visitChildren(ctx);
00109     }
00110
00111     virtual std::any visitReturn(Sysy22Parser::ReturnContext *ctx) override {
00112         return visitChildren(ctx);
00113     }
00114
00115     virtual std::any visitExp(Sysy22Parser::ExpContext *ctx) override {
00116         return visitChildren(ctx);
00117     }
00118
00119     virtual std::any visitCond(Sysy22Parser::CondContext *ctx) override {
00120         return visitChildren(ctx);
00121     }
00122
00123     virtual std::any visitLVal(Sysy22Parser::LValContext *ctx) override {
00124         return visitChildren(ctx);
00125     }
00126
00127     virtual std::any visitPrimaryExp_(Sysy22Parser::PrimaryExp_Context *ctx) override {
00128         return visitChildren(ctx);
00129     }
00130
00131     virtual std::any visitLValExpr(Sysy22Parser::LValExprContext *ctx) override {
00132         return visitChildren(ctx);
00133     }
00134
00135     virtual std::any visitIntConst(Sysy22Parser::IntConstContext *ctx) override {
00136         return visitChildren(ctx);
00137     }
00138
00139     virtual std::any visitFloatConst(Sysy22Parser::FloatConstContext *ctx) override {
00140         return visitChildren(ctx);
00141     }
00142
00143     virtual std::any visitNumber(Sysy22Parser::NumberContext *ctx) override {
00144         return visitChildren(ctx);
00145     }
00146
00147     virtual std::any visitUnaryExp_(Sysy22Parser::UnaryExp_Context *ctx) override {
00148         return visitChildren(ctx);
00149     }
00150
00151     virtual std::any visitCall(Sysy22Parser::CallContext *ctx) override {
00152         return visitChildren(ctx);
00153     }
00154
00155     virtual std::any visitUnaryAdd(Sysy22Parser::UnaryAddContext *ctx) override {
00156         return visitChildren(ctx);
00157     }
00158
00159     virtual std::any visitUnarySub(Sysy22Parser::UnarySubContext *ctx) override {
00160         return visitChildren(ctx);
00161     }
00162
00163     virtual std::any visitNot(Sysy22Parser::NotContext *ctx) override {
00164         return visitChildren(ctx);
00165     }
00166
00167     virtual std::any visitStringConst(Sysy22Parser::StringConstContext *ctx) override {
00168         return visitChildren(ctx);
00169     }

```

```

00170
00171 virtual std::any visitFuncRParam(Sysy22Parser::FuncRParamContext *ctx) override {
00172     return visitChildren(ctx);
00173 }
00174
00175 virtual std::any visitFuncRParams(Sysy22Parser::FuncRParamsContext *ctx) override {
00176     return visitChildren(ctx);
00177 }
00178
00179 virtual std::any visitDiv(Sysy22Parser::DivContext *ctx) override {
00180     return visitChildren(ctx);
00181 }
00182
00183 virtual std::any visitMod(Sysy22Parser::ModContext *ctx) override {
00184     return visitChildren(ctx);
00185 }
00186
00187 virtual std::any visitMul(Sysy22Parser::MulContext *ctx) override {
00188     return visitChildren(ctx);
00189 }
00190
00191 virtual std::any visitMulExp_(Sysy22Parser::MulExp_Context *ctx) override {
00192     return visitChildren(ctx);
00193 }
00194
00195 virtual std::any visitAddExp_(Sysy22Parser::AddExp_Context *ctx) override {
00196     return visitChildren(ctx);
00197 }
00198
00199 virtual std::any visitAdd(Sysy22Parser::AddContext *ctx) override {
00200     return visitChildren(ctx);
00201 }
00202
00203 virtual std::any visitSub(Sysy22Parser::SubContext *ctx) override {
00204     return visitChildren(ctx);
00205 }
00206
00207 virtual std::any visitGeq(Sysy22Parser::GeqContext *ctx) override {
00208     return visitChildren(ctx);
00209 }
00210
00211 virtual std::any visitLt(Sysy22Parser::LtContext *ctx) override {
00212     return visitChildren(ctx);
00213 }
00214
00215 virtual std::any visitRelExp_(Sysy22Parser::RelExp_Context *ctx) override {
00216     return visitChildren(ctx);
00217 }
00218
00219 virtual std::any visitLeq(Sysy22Parser::LeqContext *ctx) override {
00220     return visitChildren(ctx);
00221 }
00222
00223 virtual std::any visitGt(Sysy22Parser::GtContext *ctx) override {
00224     return visitChildren(ctx);
00225 }
00226
00227 virtual std::any visitNeq(Sysy22Parser::NeqContext *ctx) override {
00228     return visitChildren(ctx);
00229 }
00230
00231 virtual std::any visitEq(Sysy22Parser::EqContext *ctx) override {
00232     return visitChildren(ctx);
00233 }
00234
00235 virtual std::any visitEqExp_(Sysy22Parser::EqExp_Context *ctx) override {
00236     return visitChildren(ctx);
00237 }
00238
00239 virtual std::any visitLAndExp_(Sysy22Parser::LAndExp_Context *ctx) override {
00240     return visitChildren(ctx);
00241 }
00242
00243 virtual std::any visitAnd(Sysy22Parser::AndContext *ctx) override {
00244     return visitChildren(ctx);
00245 }
00246
00247 virtual std::any visitOr(Sysy22Parser::OrContext *ctx) override {
00248     return visitChildren(ctx);
00249 }
00250
00251 virtual std::any visitLOrExp_(Sysy22Parser::LOrExp_Context *ctx) override {
00252     return visitChildren(ctx);
00253 }
00254 };

```

7.7 src/frontend/emit_ir.cpp File Reference

7.8 src/frontend/emit_ir.hpp File Reference

7.9 emit_ir.hpp

[Go to the documentation of this file.](#)

00001

7.10 src/main.cpp File Reference

```
#include "grammar/Sysy22Lexer.h"
#include "grammar/Sysy22Parser.h"
#include "grammar/Sysy22Visitor.h"
#include "support/Any.h"
#include <antlr4-runtime/ANTLRInputStream.h>
#include <antlr4-runtime/CommonTokenStream.h>
#include <chrono>
#include <iostream>
#include <fstream>
#include <antlr4-runtime/antlr4-runtime.h>
```

Functions

- int [main](#) (int argc, char **argv)

7.10.1 Function Documentation

7.10.1.1 main()

```
int main (
    int argc,
    char ** argv)
```


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